Calculus

Write your questions and thoughts here!

4.1 Interpreting the Derivative in Context

Notes

Recall:

Slope between two points: — or ————————

Units for the Derivative:

The derivative of $f(x)$ is ————

If $f'(x) > 0$, then $f(x)$ is

If $f'(x) < 0$, then $f(x)$ is

1. Mr. Sullivan wants Mr. Brust to finish creating his packets in Algebra 2. The number of packets Mr. Brust has completed is modeled by $p(w)$, where w is measured in weeks.
   a. Interpret $p(10) = 1$ in the context of the problem.

   b. Interpret $p'(39) = 4$ in the context of the problem.

2. The rate at which Mr. Kelly is buying baseball cards per year is modeled by $b(t)$, where t is measured in years.
   a. Interpret $b(3) = 150$ in the context of the problem.

   b. Interpret $b'(4) = 10$ in the context of the problem.

4.1 Interpreting the Meaning of the Derivative in Context

Interpreting Derivatives in Context

Interpreting the meaning of values is often seen on the AP Calculus Exams. When interpreting derivative values, it is essential you include **three** pieces of information.

1. **N**_____
2. **U**_____
3. **T**_____



On the AP exam, avoid using vague phrases like "it" and "the function".

Handling Negative Rates of Change

One important caveat in interpreting derivative values is how to deal with negative rates, such as in example 2b). You can handle the negative rate in one of two ways:

1. "At $t = 20$ hours, the rate at which the temperature is **changing** at a rate of -0.5 degrees per hour per hour."

2. "At $t = 20$ hours, the rate at which the temperature is **decreasing** at a rate of (positive) 0.5 degrees per hour per hour."



*Avoid using a double negative. The explanation "At $t = 20$ hours, the rate at which the temperature is **decreasing** at a rate of -0.5 degrees per hour per hour" would actually indicate an _____ temperature.*



Ex 3: **2013 AB1 – part (a)**

On a certain workday, the rate, in tons per hour, at which unprocessed gravel arrives at a gravel processing plant is modeled by $G(t) = 90 + 45\cos\left(\frac{t^2}{18}\right)$, where t is measured in hours and $0 \le t \le 8$. At the beginning of the workday $(t = 0)$, the plant has 500 tons of unprocessed gravel. During the hours of operation, $0 \le t \le 8$, the plant processes gravel at a constant rate of 100 tons per hour.

(a) Find $G'(5)$. Using correct units, interpret your answer in the context of the problem.



Ex 4: **2007 Form B AB3 – part (a)**

The wind chill is the temperature, in degrees Fahrenheit (°F), a human feels based on the air temperature, in degrees Fahrenheit, and the wind velocity v, in miles per hour (mph). If the air temperature is 32°F, then the wind chill is given by $W(v) = 55.6 - 22.1v^{0.16}$ and is valid for $5 \le v \le 60$.

(a) Find $W'(20)$. Using correct units, explain the meaning of $W'(20)$ in terms of the wind chill.

2023 Student Samples – Use the 2023 scoring guidelines to score each student sample.

SAMPLE A

$g'(140) = -.0049$

~~The rate of flow of gasoline~~

The rate of the rate of the flow of gasoline is decreasing by $-.0049$ gallons/second/second when $t = 140$.

Point 8: 0 or 1

Point 9: 0 or 1

SAMPLE B

$g'(140) = \frac{d}{dt}\left(\frac{t}{500}\right)\cos\left(\left(\frac{t}{120}\right)^2\right)\Big|_{t=140}$

$g'(140) = -0.001$ gallons/second^2

At 140 seconds, the rate at which the rate of gasoline entering the tank is changing is equal to -0.001 gallons/second^2.

Point 8: 0 or 1

Point 9: 0 or 1

SAMPLE C

$g'(140) = -0.005$ gallons/second/second

↗ math 8

$g'(140) = -.005$ means that at $t = 140$ seconds, the rate of flow of gasoline is changing at a rate of $-.005$.

Point 8: 0 or 1

Point 9: 0 or 1

4.1 Interpreting the Derivative in Context

Calculus

Practice

For each problem, a differentiable function is given along with a definition of the variables. Interpret the values in the context of the problem.

1. The percentage grade a student receives on a test, is modeled by $G(t)$ where t is the number of hours spent studying for the test. Interpret $G'(1) = 3$.

2. Mr. Bean rides his motor scooter to work some days. His distance from home can be modeled by $d(t)$ meters where t is measured in minutes. Interpret $d'(15) = 650$.

3. The rate at which a factory produces baseball hats can be modeled by $b(t)$ where $b(t)$ is the number hats produced per hour and t is the number of hours since the factory opens. Interpret $b'(1) = 100$.

4. Mr. Brust has entered a Biggest Loser contest and is hoping to lose some of those holiday calories. His weight gain or loss can be modeled by $p(t)$, where p is measured in pounds per week and t is weeks since he started his diet. Interpret $p'(4) = -1$.

5. The number of gallons of water in a storage tank at time t, in minutes, is modeled by $w(t)$. Interpret $w'(10) = -8$.

6. The rate at which the temperature is changing is modeled by $T(h)$, where T is measured in degrees per hour and h is hours since midnight. Interpret $T'(20) = -0.5$.

7. A harbor's water depth changes with the ocean tides. The rate of change of the depth of the water is modeled by $d(t)$, where d is measured in feet per hour and t is hours. Interpret $d'(2) = -3$.

8. The height of a rocket is modeled by $h(t)$ meters where t is measured in seconds. Interpret $h'(10) = 30$.

9. The time it takes for a chemical reaction to occur can be modeled by $t(A)$, where t is the time, in minutes, and A is the catalyst used, measured in milliliters. Interpret $t'(40) = 1.7$.

10. A tire is leaking air pressure because of a small hole. The function $p(t)$ models the amount of air pressure (psi) in the tire after t minutes. Interpret $p'(3) = -2$.

No test prep for this lesson.

Calculus

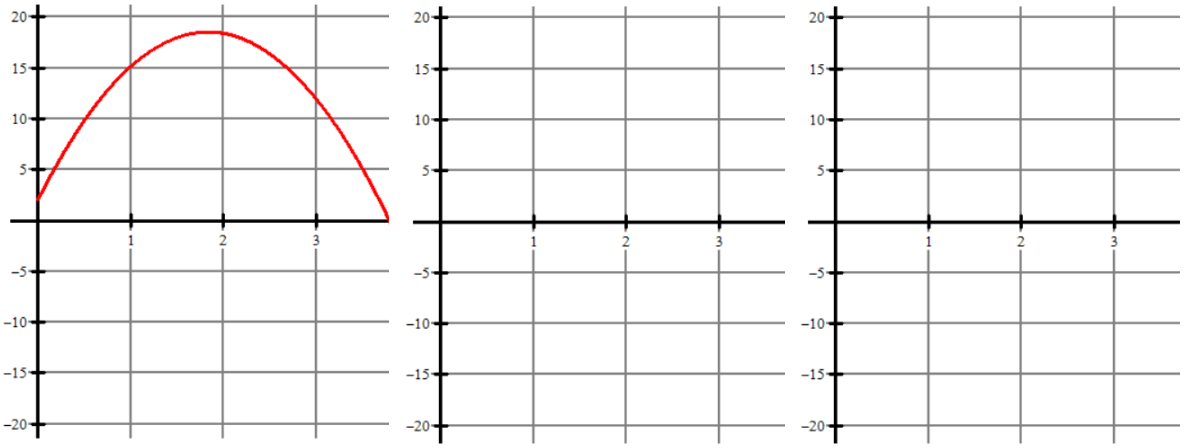
Write your questions and thoughts here!

4.2 Position, Velocity, and Acceleration

Notes

Mr. Brust is playing catch with his best friend, himself. He throws a tennis ball straight up into the air. The height of the ball is modeled by $\boldsymbol{s(t) = -4.9t^2 + 18t + 2}$ where t is time in seconds and s is the height of the ball from the ground in meters.

POSITION **VELOCITY** **ACCELERATION**



Position function: $s(t)$

Velocity function: $v(t) =$

Acceleration function: $a(t) =$

Velocity = Rate of Change of Position

$v(t) < 0$ means the particle is _____

$v(t) > 0$ means the particle is _____

$v(t) = 0$ means the particle is _____

Average velocity =

Speed =

Speeding Up or Slowing Down?

If velocity and acceleration have the **same** sign, the particle is _____

If velocity and acceleration have **different** signs, the particle is _____

t	-5	1	2	4
$v(t)$	3	-2	1	-1
$a(t)$	-4	7	0.1	-1
Conclusion				

Displacement:

Write your questions and thoughts here!

Particle Motion from an equation.

The position (x-coordinate) of a particle moving on the x-axis is modeled by the function

$$x(t) = t^3 - 4t^2 + 3 \text{ for } t \geq 0,$$

Where x is measured in cm and t is measured in minutes.

1. Find the displacement of the particle during the first 2 minutes.

2. Find the average velocity of the particle during the first 2 minutes.

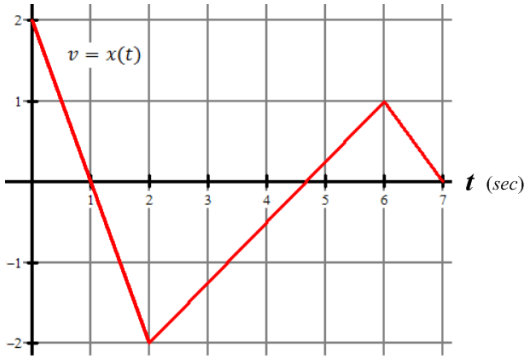
3. Find the velocity of the particle when $t = 4$.

4. Find the acceleration of the particle when $t = 4$.

5. Is the particle speeding up or slowing down at $t = 4$? Justify.

Particle Motion from a graph

The figure shows the velocity $v = x(t)$ of a particle moving on a coordinate line.



6. When is the particle moving right? Justify.

7. When is the particle moving left? Justify.

8. When is the particle's acceleration Positive? Negative? Zero?

9. When does the particle have the greatest speed?

10. When is the particle speeding up? Justify.

11. When is the particle slowing down? Justify.

Applications of Velocity - Motion Along A Line

Rest: _____
Left and Right Motion:

Left (or down): _____

Right (or up): _____

Changes Direction: _____

Velocity

When an object moves, its position changes over time. The velocity function, denoted _____, tells the _____ of change of the position function over time. Therefore to find the velocity function, _____.

Applications of Velocity – Instantaneous vs. Average Velocity

Instantaneous Velocity at $t = a$.	Average Velocity on (t_1, t_2)

Applications of Velocity – Speed

Speed and velocity are related but not the same. Speed is a _____ quantity, which means it only tells you how fast something is moving. It does not include direction. Velocity is a _____ quantity, which means it tells you both how fast and in what direction the object is moving.

Speed at $t = a$: _____





Speed is the _____ of velocity and therefore must be nonnegative.

Discovering Relationships

Ex 3: Open the GeoGebra applet.

a) Press the "start button." Observe the direction the particle moves, then fill in the blanks:

The particle moves up when $s(t)$ is _____.
The particle moves down when $s(t)$ is _____.
The particle changes direction when $s(t)$ _____.

b) Turn on the velocity function. Observe the values of the velocity function as the particle moves up and down. Then, fill in the blanks:

The particle moves up when $v(t)$ is _____.
The particle moves down when $v(t)$ is _____.
The particle changes direction when $v(t)$ _____.

Acceleration

The acceleration function, denoted _____, tells the _____ of change of the velocity function over time. Therefore to find the acceleration function, _____.

Ex 5:

A particle moves along the x-axis so that its position at time $t > 0$ is given by $x(t)$ and $\frac{dx}{dt} = -10t^4 + 9t^2 + 8t$. The acceleration of the particle is zero when $t =$

(A) 0.387 (B) 0.831 (C) 1.243 (D) 1.647 (E) 8.094

Ex 6: **2014 AB1 – part (a)**

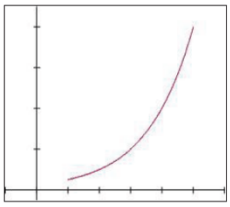
t (hours)	0	0.4	0.8	1.2	1.6	2.0	2.4
$v(t)$ (miles per hour)	0	11.8	9.5	17.2	16.3	16.8	20.1

Ruth rode her bicycle on a straight trail. She recorded her velocity $v(t)$, in miles per hour, for selected values of t over the interval $0 \le t \le 2.4$ hours, as shown in the table above. For $0 < t \le 2.4$, $v(t) > 0$.

(a) Use the data in the table to approximate Ruth's acceleration at time $t = 1.4$ hours. Show the computations that lead to your answer. Indicate units of measure.

The Relationship Between Velocity, Acceleration and Speed

Case 1: Velocity Graph



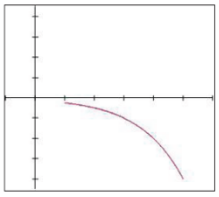
time	velocity	speed
1	1	
2	2	
3	4	
4	8	
5	16	

Sign of velocity: _____

Sign of acceleration: _____

Is speed increasing or decreasing?_____

Case 2: Velocity Graph



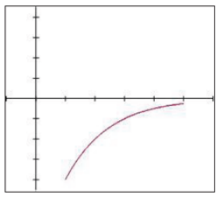
time	velocity	speed
1	-1	
2	-2	
3	-4	
4	-8	
5	-16	

Sign of velocity: _____

Sign of acceleration: _____

Is speed increasing or decreasing?_____

Case 3: Velocity Graph



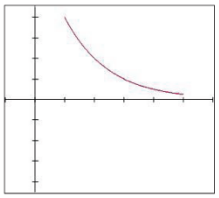
time	velocity	speed
1	-16	
2	-8	
3	-4	
4	-2	
5	-1	

Sign of velocity: _____

Sign of acceleration: _____

Is speed increasing or decreasing?_____

Case 4: Velocity Graph



time	velocity	speed
1	16	
2	8	
3	4	
4	2	
5	1	

Sign of velocity: _____

Sign of acceleration: _____

Is speed increasing or decreasing?_____

Conclusion:

- Speed is increasing when_____
- Speed is decreasing when_____

Ex 10: **2023 AB 2 – parts (a) and (b)**

Stephen swims back and forth along a straight path in a 50-meter-long pool for 90 seconds. Stephen's velocity is modeled by $v(t) = 2.38e^{-0.02t}\sin\left(\frac{\pi}{56}t\right)$, where t is measured in seconds and $v(t)$ is measured in meters per second.

(a) Find all times t in the interval $0 < t < 90$ at which Stephen changes direction. Give a reason for your answer.

(b) Find Stephen's acceleration at time $t = 60$ seconds. Show the setup for your calculations, and indicate units of measure. Is Stephen speeding up or slowing down at time $t = 60$ seconds? Give a reason for your answer.

To sum it all up…

1. "Initially" means_____=0.
2. "At rest" means_____=0.
3. If the velocity of the particle is positive, then the particle is moving to the _____.
4. If the velocity of the particle is _____, then the particle is moving to the left.
5. To find average velocity over a time interval, divide the change in _____ by the change in time.
6. Instantaneous velocity is the velocity at a single moment or _____in time.
7. If the acceleration of the particle is positive, then _____is increasing.
8. If the acceleration of the particle is _____, then the velocity is decreasing.
9. In order for a particle to change direction, the _____must change signs.
10. Speed is the _____of velocity.

4.2 Position, Velocity, and Acceleration

Calculus

Practice

1. A particle moves along a line so that its position at any time $t \geq 0$ is given by the function

$$s(t) = \frac{1}{3}t^3 - 3t^2 + 8t - 5$$

where *s* is measured in meters and *t* is measured in seconds.

a. Find the instantaneous velocity at any time *t*.

b. When is the particle at rest?

c. Find the acceleration of the particle at any time *t*.

d. What is the displacement of the particle for the first 3 seconds?

2. A ball is dropped off a 1200 ft cliff. The height of the ball over time is modeled by the function $h(t) = 1200 - 16t^2$ where h is the height of the ball and t is time in seconds.

a. Find $h'(3)$. Explain what it means.

b. Find $h''(3)$. Explain what it means.

3. The position, in meters, of a body at time *t* sec is $s(t) = t^3 - 6t^2 + 9t$. Find the body's acceleration each time the velocity is zero.

4. A particle *P* moves on the number line. The graph $s = f(t)$ shows the position of *P* as a function of time *t*.

a. When is *P* moving to the left?

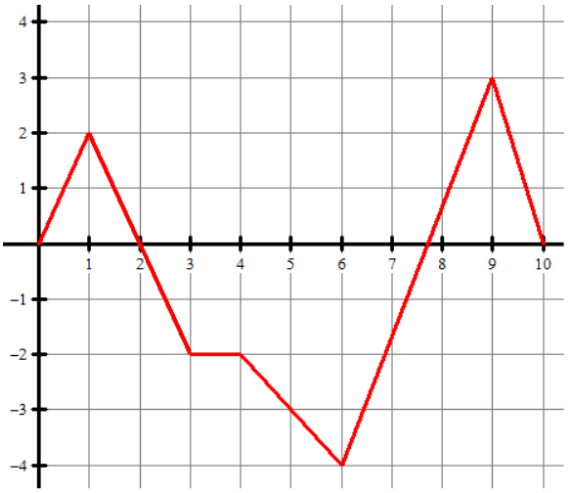
b. When is *P* moving to the right?

c. When is *P* at rest?

d. Graph the particle's velocity where defined.



5. The figure shows the velocity $v = \frac{ds}{dt} = f(t)$ of a body moving along a coordinate line in meters per second.



a. When does the body reverse direction?

b. When is the body moving at a constant speed?

c. What is the body's maximum speed?

d. What time interval(s) is the body speeding up?

6. A rock thrown vertically upward from the surface of the moon at a velocity of 32 meters per second reaches a height of $s(t) = 32t - 0.8t^2$ meters in t seconds.

a. Find the rock's velocity and acceleration as functions of time.

b. How long did it take the rock to reach its highest point?

7. The data in the table gives selected values for the velocity, in meters per minute, of a particle moving along the x-axis. The velocity v is a differentiable function of time t.

Time t	0	2	5	6	8	12
Velocity $v(t)$	-3	2	3	5	7	5

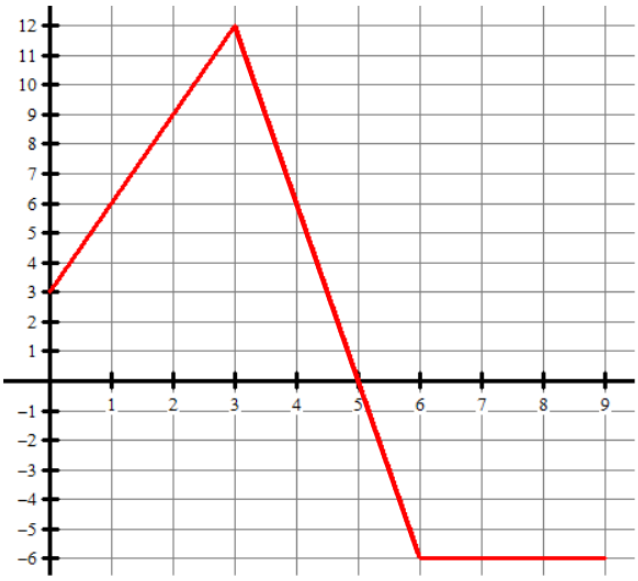
a. At $t = 0$, is the particle moving to the right or left? Justify.

b. Is there a time during the time interval $0 \le t \le 12$ minutes when the particle is at rest? Justify.

c. Use the data from the table to approximate $v'(10)$.

d. Explain the meaning of $v'(10)$ in terms of the particle motion.

8. The graph represents the velocity, in feet per second, of a particle moving along the x-axis over the time interval from $t = 0$ to $t = 9$ seconds.



a. At $t = 4$, is the particle moving to the right or left? Justify.

b. Over what time interval is the particle moving left? Justify.

c. At $t = 4$, is the acceleration positive or negative? Justify.

d. What is the average acceleration of the particle over the interval $2 \le t \le 4$? Show the computations and label your answer.

e. At what time t in the given interval is the particle furthest to the right. Justify.

---

9. A particle moves along the x-axis so that at time t its position is given by

$$x(t) = t^3 - 6t^2 + 9t + 11$$

where t is measured in seconds and x is measured in feet.

a. At $t = 0$, is the particle moving to the right or left? Justify.

b. At $t = 1$, is the velocity of the particle increasing or decreasing? Justify.

c. What is the displacement over the first 6 seconds?

No test prep. These questions are already like exam type problems.

Calculus

4.3 Rates of Change Other Than Motion

Notes

Write your questions and thoughts here!

Increasing or Decreasing?

To know if something is increasing or decreasing, check the _____

Height is *increasing* if _____

Velocity is *decreasing* if _____

Mr. Brust's IQ is *increasing* if _____

😊 is *decreasing* if _____

Recall: Derivative on a calculator.

Find $f'(3)$ if $f(x) = 5^{\sin x}$

Is the function already a rate of change?

- If $f(x)$ is the bunny population after x years, than what is $f'(x)$?

- If $f(x)$ is the rate at which a bunny population increases (bunnies per year), than what is $f'(x)$?

Rate of Change from a Table

t (years)	0	10	20	30
$P(t)$ (people)	100	120	150	200

Estimate $P'(15)$

Estimate $P'(20)$

4.3 Rates of Change in Applied Contexts Other Than Motion

Ex 2: **2014 IPE – AB2 part (a)**

A store is having a 12-hour sale. The total number of shoppers who have entered the store t hours after the sale begins is modeled by the function S defined by $S(t) = 0.5t^4 - 16t^3 + 144t^2$ for $0 \le t \le 12$. At time $t = 0$, when the sale begins, there are no shoppers in the store.

(a) At what rate are shoppers entering the store 3 hours after the start of the sale?

Ex 3: **2004 (Form B) – AB2 parts (a) and (b)**

For $0 \le t \le 31$, the rate of change of the number of mosquitoes on Tropical Island at time t days is modeled by $R(t) = 5\sqrt{t}\cos\left(\frac{t}{5}\right)$ mosquitoes per day. There are 1000 mosquitoes on Tropical Island at time $t = 0$.

(a) Show that the number of mosquitoes is increasing at time $t = 6$.

(b) At time $t = 6$, is the number of mosquitoes increasing at an increasing rate, or is the number of mosquitoes increasing at a decreasing rate? Give a reason for your answer.

Be very careful when approaching these problems. Stop and think BEFORE finding a derivative! Use the _____ of the given function(s) as a clue.

Ex 4: **2022 AB1 – part (c)**

From 5 A.M. to 10 A.M., the rate at which vehicles arrive at a certain toll plaza is given by $A(t) = 450\sqrt{\sin(0.62t)}$, where t is the number of hours after 5 A.M. and $A(t)$ is measured in vehicles per hour. Traffic is flowing smoothly at 5 A.M. with no vehicles waiting in line.

(c) Is the rate at which vehicles arrive at the toll plaza at 6 A.M. ($t = 1$) increasing or decreasing? Give a reason for your answer.

Ex 5: **2019 AB1 – part (d)**



Fish enter a lake at a rate modeled by the function E given by $E(t) = 20 + 15\sin\left(\frac{\pi t}{6}\right)$. Fish leave the lake at a rate modeled by the function L given by $L(t) = 4 + 2^{0.1t^2}$. Both $E(t)$ and $L(t)$ are measured in fish per hour, and t is measured in hours since midnight ($t = 0$).

(d) Is the rate of change in the number of fish in the lake increasing or decreasing at 5 A.M. ($t = 5$) ? Explain your reasoning.

Ex 6: **2017 AB2 – parts (b) and (c)**



When a certain grocery store opens, it has 50 pounds of bananas on a display table. Customers remove bananas from the display table at a rate modeled by

$$f(t) = 10 + (0.8t)\sin\left(\frac{t^3}{100}\right) \text{ for } 0 < t \le 12,$$

where $f(t)$ is measured in pounds per hour and t is the number of hours after the store opened. After the store has been open for three hours, store employees add bananas to the display table at a rate modeled by

$$g(t) = 3 + 2.4\ln\left(t^2 + 2t\right) \text{ for } 3 < t \le 12,$$

where $g(t)$ is measured in pounds per hour and t is the number of hours after the store opened.

(b) Find $f'(7)$. Using correct units, explain the meaning of $f'(7)$ in the context of the problem.

(c) Is the number of pounds of bananas on the display table increasing or decreasing at time $t = 5$? Give a reason for your answer.

Ex 7:

People are entering a building at a rate modeled by $f(t)$ people per hour and exiting the building at a rate modeled by $g(t)$ people per hour, where t is measured in hours. The functions f and g are nonnegative and differentiable for all times t. Which of the following inequalities indicates that the rate of change of the number of people in the building is increasing at time t ?

(A) $f(t) > 0$

(B) $f'(t) > 0$

(C) $f(t) - g(t) > 0$

(D) $f'(t) - g'(t) > 0$

4.3 Rates of Change Other Than Motion

Practice

Calculus

1. A store is having a 12-hour sale. The total number of shoppers who have entered the store t hours after the sale begins is modeled by the function E defined by $E(t) = 0.3t^4 - 14t^3 + 110t^2$ for $0 \le t \le 12$. At what rate are shoppers entering the store 5 hours after the start of the sale?

---

2. The function $t = f(P)$ models the time, in days, for a small pond to evaporate as a function of the size P of the pond, measured in liters. What are the units for $f''(P)$?

3.

t (days)	$W(t)$ (°F)
0	58
5	60
10	61
15	64
20	68

The temperature, in degrees Fahrenheit (°F), of a lake is a differentiable function W of time t. The table above shows the water temperature as recorded every 5 days over a 20-day period. Use the data from the table to find an approximation for $W'(10)$. Show the computations that lead to your answer. Indicate units of measure.

---

4. The function $D(t) = 13 - 2.7\cos\left(\frac{\pi}{4}t\right)$ models the depth, in feet, of water t hours after 6 A.M. Find the instantaneous rate of change of the depth of the water at 9 A.M. Use appropriate units.

---

5. The rate of consumption of liquid chocolate in Switzerland is given by $R(t) = ce^{kt}$, where R is measured in millions of gallons per year and t is measured in years from the beginning of 1990. The consumption rate doubles every 9 years and the consumption rate at the beginning of 1990 was 2 million gallons per year. Find c and k.

6.

t (minutes)	0	10	20	30
$W(t)$ (°F)	104	90	79	72

The temperature of water in a bathtub at time t is modeled by a strictly decreasing, twice-differentiable function W, where $W(t)$ is measured in degrees Fahrenheit and t is measured in minutes. The water is cooling for 30 minutes, beginning at time $t = 0$. Values of $W(t)$ at selected times t are given in the table above. Use the data in the table to estimate $W'(15)$. Show the computations that lead to your answer. Using correct units, interpret the meaning of your answer in the context of this problem.

---

7. The rate of decay, in grams per minute, of a radioactive substance is a differentiable, decreasing function R of time, t, in minutes. The table below shows the decay rate as recorded every 4 minutes over a 24-minute period.

t (minutes)	0	4	8	12	16	20	24
$R(t)$ (grams per minute)	320	221	130	82	39	22	11

a. Approximate $R'(12)$. Show the computations that lead to your answer. Indicate units of measure.

b. A physicist proposes the function $G(t) = 320(0.882)^t$ as a model for the rate of decay of the radioactive substance, using the same units as $R(t)$. Find $G'(12)$. Using appropriate units, explain the meaning of your answer in terms of the decay rate of the substance.

---

8. For time $t \geq 0$ hours, let $r(t) = 50\left(1 - e^{-0.7t^2}\right)$ represent the speed, in miles per hour, at which a car travels along a straight road. Find $r'(2)$. Indicate units of measure.

4.3 Rates of Change Other Than Motion

Test Prep

9. People are entering a museum at a rate modeled by $f(t)$ people per hour and exiting the building at a rate modeled by $g(t)$ people per hour, where t is measured in hours. The functions f and g are nonnegative and differentiable for all times t. Which of the following inequalities indicates that the rate of change of the number of people in the building is decreasing at time t?

(A) $g(t) < 0$ (B) $g'(t) < 0$ (C) $f(t) - g(t) < 0$ (D) $f'(t) - g'(t) < 0$

---

10. **Calculator Active.** The rate of $R(t)$ of a certain item produced in a factory is given by

$$R(t) = 4000 + 48(t-3) - 4(t-3)^3$$

where t is the number of hours since the beginning of the workday at 8:00 a.m. At what time is the rate of production increasing most rapidly?

(A) 8:00 am (B) 10:00 am (C) 11:00 am (D) 1:00 pm

Calculus

4.4 Introduction to Related Rates

Notes

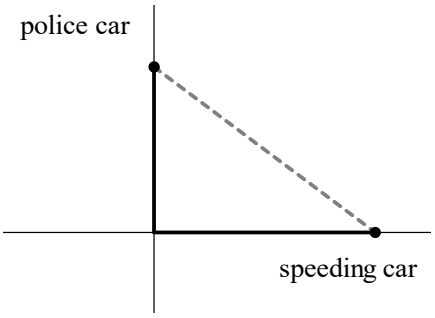
Write your questions and thoughts here!

Many variables have a familiar relationship. Or in other words, a way variables are "related" to each other. A great example is the Pythagorean Theorem.

Let's say a triangle's dimensions are changing. This means a dimension (a, b, or c) has a rate of change, and this rate is "related" to another dimension's rate of change.

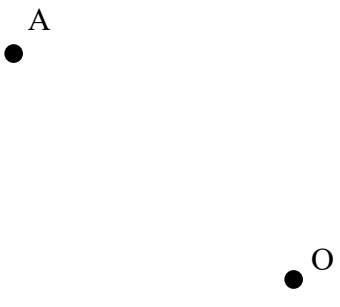
Differentiate your relationship with respect to one specific variable. This is like implicit differentiation. Usually it is with respect to time, but not always.

1. A police car, approaching a right-angled intersection from the north is chasing a speeding car that has turned the corner and is now moving straight east. Set up a relationship, then find an equation that shows the related rates of the vehicles with respect to time.



<u>Some dimensions are constants.</u>

An airplane (point A) is flying on a horizontal path that will take it directly over an observer (point O). The airplane maintains a constant altitude. Relate the rates of change of the distance between the observer and the airplane and the horizontal distance between the two.



4.4 Introduction to Related Rates

Rates of Change

We have previously learned that $\frac{dy}{dx}$ is the _____ of change of y with respect to x. To find a derivative with respect to another variable, such as t, use the _____ rule.

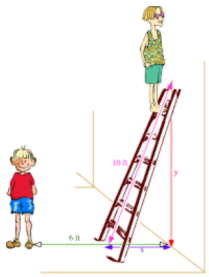
With respect to x	With Respect to t
$\frac{d}{dx}(x^2) =$ _____	$\frac{d}{dt}(x^2) =$ _____
$\frac{d}{dx}(\sqrt{y}) =$ _____	$\frac{d}{dt}(\sqrt{y}) =$ _____
$\frac{d}{dx}(\cos y) =$ _____	$\frac{d}{dt}(\cos y) =$ _____
$\frac{d}{dx}(\pi r^2) =$ _____	$\frac{d}{dt}(\pi r^2) =$ _____

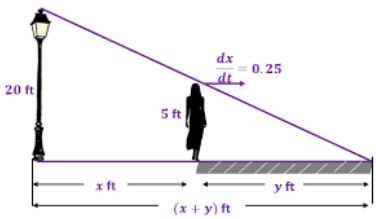
Ex 1: Differentiate with respect to time, t.

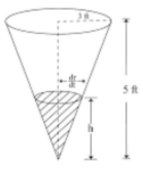
a) $a^2 + b^2 = c^2$	b) $P = 2l + 2h$
c) $V = \pi r^2 h$	d) $\cos\theta = \frac{x}{3}$

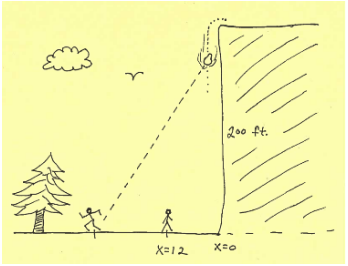
What are related rates?

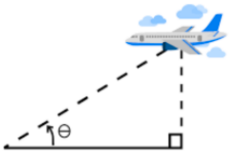
In differential calculus, related rates are two or more rates that are simultaneously changing with respect to _____.











Positive and Negative Rates

- If a quantity is increasing with respect to time, its rate of change is _____.
- If a quantity is decreasing with respect to time, its rate of change is _____.

Lesson 4.4 – Introduction to Related Rates

QuickNotes

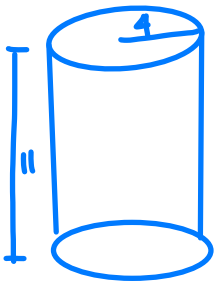
Related Rate problems have multiple variables changing with respect to the same ind. variable (usually time)

1. Draw a picture.
2. Write an equation that relates all the variables in the problem (often a volume formula or Pythag. thm)
3. Determine which variables are changing + which are constant.
4. Differentiate both sides. (Don't forget chain rule!)
5. Substitute known values + solve for the desired quantity.

* Determine based on context if the given rates are positive or negative.

Check Your Understanding

1. A cylinder filled with water has a radius of 4 feet and a height of 11 feet. Water is being drained from the container such that the height is decreasing at 0.1 feet per second. How fast is the volume of water in the container changing? $dh/dt < 0$ $\frac{dV}{dt} = ?$



   a. Write an equation that relates all the variables in the problem.

$$V = \pi r^2 h$$

   b. Which variables are remaining constant? Replace these values into your equation.

$r = 4$ $\quad V = \pi \cdot 4^2 \cdot h = 16\pi h$

   c. Write an equation for $\frac{dV}{dt}$. Don't forget the chain rule!

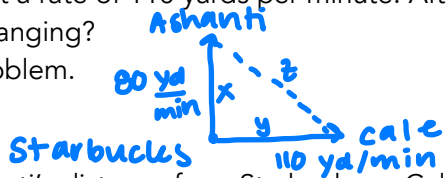
$$\frac{dV}{dt} = \frac{dh}{dt} \cdot 16\pi$$

   d. Plug in the known values and solve for the quantity you are after.

$$\frac{dV}{dt} = 16\pi(-0.1) = -1.6\pi \text{ ft}^3/\text{s}$$

2. Ashanti and Cale met at Starbucks and are now returning home. Ashanti heads north at a rate of 80 yards per minute and Cale heads east at a rate of 110 yards per minute. After 15 minutes, at what rate is the distance between them changing?

   a. Draw a picture to represent the problem.



   b. Write an equation that relates Ashanti's distance from Starbucks, *x*, Cale's distance from Starbucks, *y*, and their distance from each other, *z*.

$$x^2 + y^2 = z^2$$

$\frac{dx}{dt} = 80$ yd/min

$\frac{dy}{dt} = 110$ yd/min

   c. Take the derivative of both sides.

$$2x\frac{dx}{dt} + 2y \cdot \frac{dy}{dt} = 2z \cdot \frac{dz}{dt}$$

$$\frac{2x\frac{dx}{dt} + 2y\frac{dy}{dt}}{2z} = \frac{dz}{dt}$$

   d. Plug in the known values and solve for the quantity you are after.

After 15 min, $x = 80(15) = 1200$ yd $\quad y = 110(15) = 1650$ yd

$$z = \sqrt{1200^2 + 1650^2} = 2040.221 \text{ yd}$$

$$\frac{dz}{dt} = 136.015 \text{ yd/min}$$

Ex 6: **2024 AB5 – part (d)**

(d) For time $t \geq 0$, a particle is moving along another curve defined by the equation $y^3 + 2xy = 24$. At the instant the particle is at the point $(4, 2)$, the y-coordinate of the particle's position is decreasing at a rate of 2 units per second. At that instant, what is the rate of change of the x-coordinate of the particle's position with respect to time?

4.4 Introduction to Related Rates

Calculus

Practice

Find a relationship between the given rates of change by doing the following.

1. **Set up a relationship using variables for the situation.**
2. **Differentiate with respect to time t.**

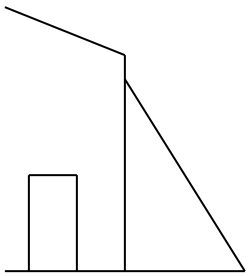
1. An ice cube is melting. Relate the volume's rate of change with the edges' rate of change.

2. The width and length of a rectangle are increasing. Relate these rates of change with the rate of change of the area of the rectangle.

3. A spherical balloon is expanding. Relate the rate of change of surface area with the rate of change of the radius of the balloon. $A = 4\pi r^2$.

4. The water level is dropping in a cylindrical tank because of a small leak in the tank. Relate the rates of change between the water level and the volume of the water. The volume is modeled by $V = \pi r^2 h$.

5. Mr. Brust is using a ladder to paint his house. The ladder is leaning against the house when Mr. Sullivan decides to pull the base of the ladder away from the house. Set up a relationship of the rates of change of how high the top of the ladder is and the distance from the bottom of the ladder to the house.



6. A rocket is rising vertically. An observer on the ground is standing a certain distance from the rocket's launch point. As the observer watches the rocket, the angle of elevation is increasing. Relate the rates of change of the angle of elevation with the speed of the rocket.

7. A boat is being pulled toward a dock by a rope attached to its bow through a pulley on the dock. The pulley is higher than the boat's bow by several feet. Relate the rate that the rope is hauled in with how fast the boat is approaching the dock.

4.4 Introduction to Related Rates

Test Prep

8. Let a and b be functions of time t such that the sum of a and three times b is constant. Which of the following equations describes the relationship between the rate of change of a with respect to time and the rate of change of b with respect to time?

(A) $\frac{da}{dt} = 3\frac{db}{dt}$

(B) $\frac{da}{dt} = -3\frac{db}{dt}$

(C) $3\frac{da}{dt} + \frac{db}{dt} = 0$

(D) $\frac{da}{dt} + 3\frac{db}{dt} = F$, where F is a function of t.

---

9. The number of minutes M that it takes to make a calculus lesson and the number of lessons L that are made per week satisfy the relationship $L = \frac{k}{M}$, where k is a constant. Which of the following best describes the relationship between the rate of change, with respect to time t, of L and the rate of change, with respect to time t, of M?

(A) $\frac{dL}{dt} = \frac{k}{\left(\frac{dM}{dt}\right)}$

(B) $\frac{dL}{dt} = \frac{-k}{\left(\frac{dM}{dt}\right)}$

(C) $\frac{dL}{dt} = \frac{k}{M^2}\left(\frac{dM}{dt}\right)$

(D) $\frac{dL}{dt} = \frac{-k}{M^2}\left(\frac{dM}{dt}\right)$

Calculus

4.5 Solving Related Rates

Notes

Write your questions and thoughts here!

1. If $y^2 = 3x^4 + 6x$, find $\frac{dy}{dt}$ when $x = 1$, and $\frac{dx}{dt} = 2$.

Guidelines to solving related rate problems

1. Draw a picture.
2. Make a list of all known and unknown rates and quantities.
3. Relate the variables in an equation.
4. Differentiate with respect to time.
5. Substitute the known quantities and rates and solve.

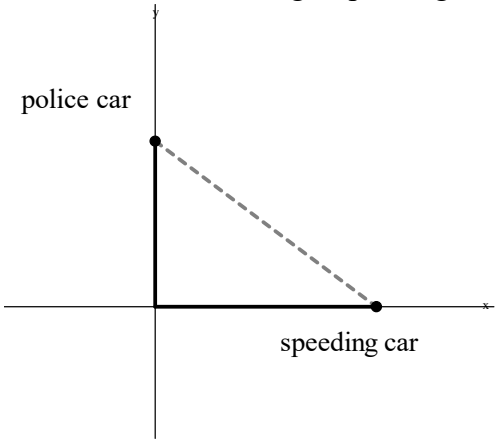
IMPORTANT: Substituting a non-constant quantity before differentiating is not allowed!

Rectangle Example

The width of a rectangle is increasing at a rate of 2 cm/sec and its length is increasing at a rate of 3 cm/sec. At what rate is the area of the rectangle increasing when its width is 4 cm and its length is 5 cm?

Triangle Example

A police car, approaching a right-angled intersection from the north is chasing a speeding car that has turned the corner and is now moving straight east. When the police car is 0.6 miles north of the intersection and the car is 0.8 miles to the east, the police determine with a radar gun that the distance between them and the car is increasing at 20 mph. If the police car is moving at 60 mph at the instant of measurement, what is the speed of the car?



4.5 Solving Related Rates

Calculus

Practice

1. If $y = 3x^4 + 6x$, find $\frac{dy}{dt}$ when $x = 1$, and $\frac{dx}{dt} = -3$.

2. If $g = 5h - h^5$, find $\frac{dg}{dt}$ when $h = 2$, and $\frac{dh}{dt} = 3$.

3. If $x^2 + y^2 = z^2$, find $\frac{dy}{dt}$ when $x = 3$, $y = 4$, $\frac{dx}{dt} = -1$, and $\frac{dz}{dt} = 5$.

4. If $A = \frac{1}{2}bh$, find $\frac{dA}{dt}$ when $b = 7$, $h = 6$, $\frac{db}{dt} = 2$, and $\frac{dh}{dt} = -3$.

5. An ice cube is melting at a rate of 5 cubic cm per hour. At what rate is the edge of the cube changing when the edge of the cube is 3 cm.

Answer: $-\frac{5}{27}$ cm/hour

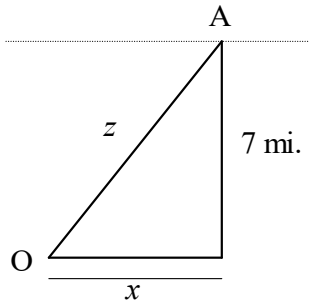
6. A circular pool of water is expanding at the rate of 16π in^2 / sec. At what rate is the radius expanding when the radius is 4 inches?

Answer: 2 inches/sec

7. A spherical balloon is expanding at a rate of 60π in^3 / sec. How fast is the surface area of the balloon expanding when the radius of the balloon is 4 inches? $V = \frac{4}{3}\pi r^3$ and $A = 4\pi r^2$.

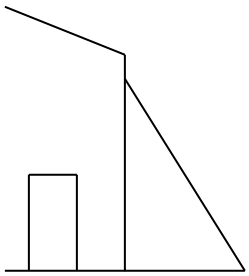
Answer: 30π in^2/sec

8. An airplane (pt. A) is flying 600 mph on a horizontal path that will take it directly over an observer (pt. O). The airplane maintains a constant altitude of 7 miles (see figure). What is the rate of change of the distance between the observer and the airplane when $x = 5$ miles?



Answer: $-\frac{3000}{\sqrt{74}}$ mph

9. Mr. Brust is using a ladder to paint his house. The 17-ft ladder is leaning against the house when Mr. Kelly decides to pull the base of the ladder away from the house at a rate of 3 ft./sec. How fast is the top of the ladder moving down the side of the house when it is 8 ft. above the ground? Indicate units of measure.



Answer: $-\frac{45}{8}$ feet/sec

10. A boat is being pulled toward a dock by a rope attached to its bow through a pulley on the dock. The pulley is 7 feet higher than the boat's bow. If the rope is hauled in at a rate of 4 feet/sec, how fast is the boat approaching the dock when 25 feet of rope is out?

Answer: $\frac{25}{6}$ feet/sec

4.5 Solving Related Rates

Test Prep

11. The base of a triangle is decreasing at a constant rate of 0.2 cm/sec and the height is increasing at 0.1 cm/sec. If the area is increasing, which answer best describes the constraints on the height *h* at the instant when the base is 3 centimeters?

(a) $h > 3$ (b) $h < 1$ (c) $h > 1.5$ (d) $h < 1.5$ (e) $h > 2$

Calculus

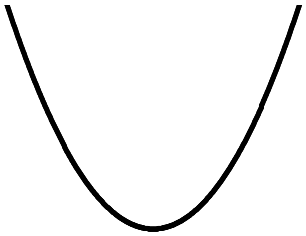
Write your questions and thoughts here!

4.6 Approximating with Local Linearity

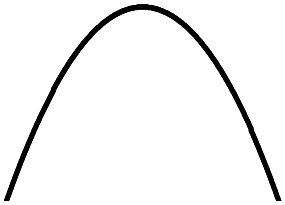
Notes

The tangent line of the function $f(x)$ at $x = a$ can give you an approximate value of $f(x)$ for points close to $x = a$.

Concave UP with a Tangent Line



Concave DOWN with a Tangent Line



1. f is concave up on its domain with $f(4) = 5$ and $f'(4) = 3$.
   a. What is the estimate for $f(3.8)$ using the local linear approximation for f at $x = 4$?

   b. Is it an underestimate or overestimate? Explain.

2. The function $f(x) = 5x - 2x^3 - 2$ is concave down at $x = 1$?.
   a. Find the tangent line of f at $x = 1$.

   b. What is the estimate for $f(1.1)$ using the local linear approximation for f at $x = 1$?

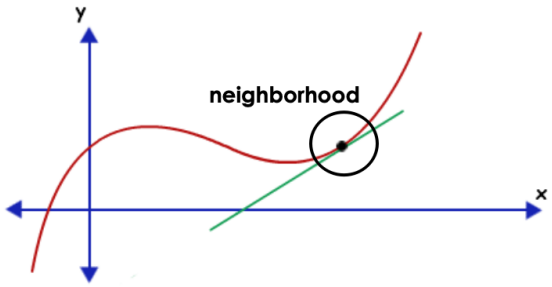
   c. Is it an underestimate or overestimate? Explain.

3. Consider the differential equation $\frac{dy}{dx} = e^y(2x^2 - 5x)$. Let $y = f(x)$ be the particular solution to the differential equation with the initial condition $f(2) = 0$.
   a. Write an equation for the line tangent to the graph of f at the point $(2,0)$.

   b. Use the tangent line to approximate $f(2.2)$.

What is a Linear Approximation and Why?

A linear approximation is also known as a "tangent line approximation." Linear approximation is used to make quick and simple _____ by approximating a function's value near a specific point using a _____. Notice on the graph below that the tangent line takes on similar values to the function *near* the _____.



To find a linear approximation:

1. Write an equation of a tangent line. On the AP exam you will be given an x –value or a point of tangency.
2. Find the desired approximation by substituting into the tangent line.

Ex 2:

Let f be a differentiable function such that $f(2) = 4$ and $f'(2) = -\frac{1}{2}$. What is the approximation for $f(2.1)$ found by using the line tangent to the graph of f at $x = 2$?

(A) 2.95 (B) 3.95 (C) 4.05 (D) 4.1

Ex 3:

The approximate value of $y = \sqrt{4 + \sin x}$ at $x = 0.12$, obtained from the tangent to the graph at $x = 0$, is

(A) 2.00 (B) 2.03 (C) 2.06 (D) 2.12 (E) 2.24

Ex 5: **2025 AB6 – part (b)**

Consider the curve G defined by the equation $y^3 - y^2 - y + \frac{1}{4}x^2 = 0$.

A. Show that $\frac{dy}{dx} = \frac{-x}{2(3y^2 - 2y - 1)}$.

B. There is a point P on the curve G near $(2, -1)$ with x-coordinate 1.6. Use the line tangent to the curve at $(2, -1)$ to approximate the y-coordinate of point P.

scoring stats!

Guess the mean: _____/9

Ex 6: **2023 AB3 – part (c)**

A bottle of milk is taken out of a refrigerator and placed in a pan of hot water to be warmed. The increasing function M models the temperature of the milk at time t, where $M(t)$ is measured in degrees Celsius (°C) and t is the number of minutes since the bottle was placed in the pan. M satisfies the differential equation $\frac{dM}{dt} = \frac{1}{4}(40 - M)$. At time $t = 0$, the temperature of the milk is 5°C. It can be shown that $M(t) < 40$ for all values of t.

(b) Use the line tangent to the graph of M at $t = 0$ to approximate $M(2)$, the temperature of the milk at time $t = 2$ minutes.

scoring stats!

Guess the mean: _____/9

Over and Underestimates

With a graphing calculator it is easy to determine if an approximation is an over or underestimate. We can also make these determinations WITHOUT technology!

Ex 8: Open the GeoGebra applet.

a) Move the green point along the curve. Fill in the blanks:

- The approximation is an **overestimate** when the curve is _____.
- The approximation is an **underestimate** when the curve is _____.

b) What happens to the approximation as the purple point moves away from the point of tangency?

4.6 Approximating with Local Linearity

Calculus

Practice

For each differential equation, let $y = f(x)$ be the particular solution to the differential equation with the given initial condition.

1. $\frac{dy}{dx} = (5 - y)\sin x$ and $f\left(\frac{\pi}{2}\right) = 2$.
   a. Write an equation for the line tangent to the graph of f at the point $\left(\frac{\pi}{2}, 2\right)$.

   b. Use the tangent line to approximate $f(1.5)$.

2. $\frac{dy}{dx} = -\frac{4x}{y}$ and $f(1) = 3$.
   a. Write an equation for the line tangent to the graph of f at the point $(1,3)$.

   b. Use the tangent line to approximate $f(1.1)$.

Answer the questions for each function listed.

3. $f(x) = 2\cos x + 1$ is concave down on $\left[0, \frac{\pi}{2}\right]$.
   a. What is the estimate for $f(1)$ using the local linear approximation for f at $x = \frac{\pi}{2}$? Give an exact answer (no rounding).

   b. Is it an underestimate or overestimate? Explain.

4. $f(x) = \frac{e^{2x}}{x+1}$ is concave up on $x > -1$.
   a. What is the estimate for $f(0.1)$ using the local linear approximation for f at $x = 0$?

   b. Is it an underestimate or overestimate? Explain.

5. $f(x) = -\sqrt{4-x}$ is concave up on its domain.
   a. What is the estimate for $f(1.9)$ using the local linear approximation for f at $x = 2$? Round to three decimal places.

   b. Is it an underestimate or overestimate? Explain.

6. f is concave down and $f(3) = -1$ and $f'(3) = 2$.
   a. What is the estimate for $f(3.2)$ using the local linear approximation for f at $x = 3$?

   b. Is it an underestimate or overestimate? Explain.

7. f is concave up and $f(-5) = 2$ and $f'(-5) = -1$.
   a. What is the estimate for $f(-5.1)$ using the local linear approximation for f at $x = -5$?

   b. Is it an underestimate or overestimate? Explain.

8. f is concave down and $f(2) = 1$ and $f'(2) = -3$.
   a. What is the estimate for $f(1.9)$ using the local linear approximation for f at $x = 2$?

   b. Is it an underestimate or overestimate? Explain.

4.6 Approximating with Local Linearity

Test Prep

9. Let f be the function given by $f(x) = 3x^2 - 4x + 2$. The tangent line to the graph of f at $x = 1$ is used to approximate values of $f(x)$. Which of the following is the smallest value of x for which the error resulting from this tangent line approximation is more than 0.5?
   [*Hint for your calculator use*: Create a table to compare values of two functions.]

   (A) 1.3 (B) 1.4 (C) 1.5 (D) 1.6 (E) 1.7

10. The depth of snow in a field is given by the twice-differentiable function S for $0 \le t \le 12$, where $S(t)$ is measured in centimeters and time t is measured in hours. Values of $S'(t)$, the derivative of S, at selected values of time t are shown in the table above. It is known that the graph of S is concave down for $0 \le t \le 12$.

t (hours)	0	1	4	9	12
$S'(t)$ (centimeters per hour)	1.8	2.4	2.0	1.6	1.3

a. Use the data in the table to approximate $S''(10)$. Show the computations that lead to your answer. Using correct units, explain the meaning of $S''(10)$ in the context of the problem.

b. Is there a time t, for $0 \le t \le 12$, at which the depth of snow is changing at a rate of 1.5 centimeters per hour? Justify your answer?

c. At time $t = 4$, the depth of snow is 28 centimeters. Use the line tangent to the graph of S at $t = 4$ to approximate the depth of the snow at time $t = 6$. Is the approximation an underestimate or an overestimate of the actual depth of snow at time $t = 6$? Justify your answer.

Calculus

4.7 L'Hospital's Rule

Notes

Write your questions and thoughts here!

Recall: When evaluating limits, first try direct substitution! $\lim_{x\to 3}\frac{2x-5}{x}=$

1. $\lim_{x\to 2}\frac{x^2-7x+10}{x-2}=$

L'Hospital's Rule:

Suppose $f(a)=0$ and $g(a)=0$ and $\lim_{x\to a}\frac{f(x)}{g(x)}=\frac{0}{0}$ or $\frac{\infty}{\infty}$. L'Hopital's Rule allows you to apply the following:

Evaluate each limit. Use L'Hospital's when possible.

2. $\lim_{x\to 2}\frac{x-2}{3x^3-6x^2+x-2}$

3. $\lim_{x\to 0}\frac{\sin(6x)}{x}$

4. $\lim_{x\to 0}\frac{1-\cos(x)}{x^2}$

5. $\lim_{x\to \infty}\frac{2x^2}{e^{2x}}$

L'HOSPITAL'S IS NOT THE QUOTIENT RULE!!

6. $\frac{d}{dx}\frac{\sin(6x)}{x}$

Indeterminate Forms

In calculus, an indeterminate form is an expression in a limit that has an unclear value because the limits of the numerator and denominator do not, by themselves, provide enough information to determine the limit of the entire expression.

In AP Calculus AB you will focus on two indeterminate forms: _____ and _____ . But don't worry, you will continue your study of indeterminate forms in AP Calculus BC and beyond!

In Unit 1 you learned how to handle these indeterminate forms using a variety of techniques like factoring, multiplying by a conjugate pair and using special trigonometric limits. In this lesson, you will learn a new technique in resolving an indeterminate form: L'Hopital's Rule.

L'Hopital's Rule

Let f and g be functions that are differentiable on an open interval (a, b) containing c, except possibly at c itself. Assume $g'(x) \neq 0$ for all x in (a, b), except possibly at c itself. If the limit as x approaches c produces an indeterminate form $\frac{0}{0}$ or $\pm\frac{\infty}{\infty}$, then

$$\lim_{x\to c}\frac{f(x)}{g(x)} = _____$$



L'Hopital's Rule can be applied repeatedly if $\lim_{x\to c}\frac{f'(x)}{g'(x)}$ also produces an indeterminate form!

Ex 1: Determine if each limit is a candidate for L'Hopital's Rule? Explain.

a) $\lim_{x\to 1}\frac{x^2+x-2}{4x-4}$	b) $\lim_{x\to\infty}\frac{x^2+x-2}{4x^2-4}$	c) $\lim_{x\to 0}\frac{\sin 6x}{x}$
d) $\lim_{x\to 1}\frac{e^x}{3x}$	e) $\lim_{x\to\infty}\frac{e^x}{3x}$	f) $\lim_{x\to 0}\frac{e^x}{3x}$

Notation Notation Notation!

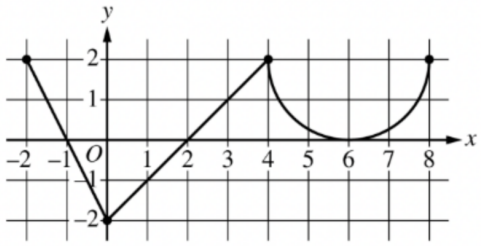
The AP Calculus exams have a high standard of communication and reasoning with theorems and definitions. To apply L'Hospital's Rule, students must first explicitly show that the conditions have been met. This requires students to examine the limits of the numerator and denominator _____ before applying L'Hospital's Rule.

For example: $\lim_{x\to 0}\frac{e^{2x}-1}{x}$

Correct Method	Incorrect Method
$\lim_{x\to 0}(e^{2x}-1)=0$ and $\lim_{x\to 0}x=0$ By L'Hopital's Rule, $\lim_{x\to 0}\frac{e^{2x}-1}{x}=\lim_{x\to 0}\frac{2e^{2x}}{1}=2$	$\lim_{x\to 0}\frac{e^{2x}-1}{x}=\frac{0}{0}$ By L'Hopital's Rule, $\lim_{x\to 0}\frac{e^{2x}-1}{x}=\lim_{x\to 0}\frac{2e^{2x}}{1}=2$

Students that link a limit with an indeterminate form, as in the incorrect method shown above, run the risk of not earning full credit on Free Response questions requiring L'Hopital's Rule.

Ex 8: **2023 AB4 – part (c)**



Graph of f'

The function f is defined on the closed interval $[-2, 8]$ and satisfies $f(2) = 1$. The graph of f', the derivative of f, consists of two line segments and a semicircle, as shown in the figure.

(c) Find the value of $\lim_{x \to 2} \dfrac{6f(x) - 3x}{x^2 - 5x + 6}$, or show that it does not exist. Justify your answer.

SCORING STATS!

Guess the mean: _____/9

Ex 9: Read the 2023 AB4 part (c) scoring guidelines and scoring notes.

Because f is differentiable at $x = 2$, f is continuous at $x = 2$, so $\lim_{x \to 2} f(x) = f(2) = 1$. $\lim_{x \to 2} (6f(x) - 3x) = 6 \cdot 1 - 3 \cdot 2 = 0$ $\lim_{x \to 2} (x^2 - 5x + 6) = 0$	Limits of numerator and denominator	**Point 4 (P4)**
Because $\lim_{x \to 2} \dfrac{6f(x) - 3x}{x^2 - 5x + 6}$ is of indeterminate form $\dfrac{0}{0}$, L'Hospital's Rule can be applied. Using L'Hospital's Rule, $\lim_{x \to 2} \dfrac{6f(x) - 3x}{x^2 - 5x + 6} = \lim_{x \to 2} \dfrac{6f'(x) - 3}{2x - 5} = \dfrac{6 \cdot 0 - 3}{2 \cdot 2 - 5} = 3$	Uses L'Hospital's Rule	**Point 5 (P5)**
	Answer	**Point 6 (P6)**

Scoring notes:

- The first point is earned by the presentation of two separate limits for the numerator and denominator.
- A response that presents a limit explicitly equal to $\dfrac{0}{0}$ does not earn the first point.
- The second point is earned by applying L'Hospital's Rule, that is, by presenting at least one correct derivative in the limit of a ratio of derivatives.
- The third point is earned for the correct answer with supporting work.

HOW DID THEY DO?

2023 Student Samples – Use the 2023 scoring guidelines for AB4 part (c) to score each student sample.

SAMPLE A

$\lim_{x\to 2} \frac{6f(x)-3x}{x^2-5x+6}$

$\lim_{x\to 2} 6f(x)-3x = 6f(2)-3(2) = 6(1)-6 = 0$

$\lim_{x\to 2} x^2-5x+6 = 4-10+6 = 0$

indeterminate form, therefore L'Hopital's Rule applies.

$= \lim_{x\to 2} \frac{6f'(x)-3}{2x-5}$

$\therefore \lim_{x\to 2} \frac{6f(x)-3x}{x^2-5x+6} = -3$

$= \frac{6f'(2)-3}{2(2)-5} = \frac{6(1)-3}{4-5} = -3$

Point 4: 0 or 1

Point 5: 0 or 1

Point 6: 0 or 1

SAMPLE B

$\lim_{x\to 2} \frac{6f(x)-3x}{x^2-5x+6} = \frac{6f(2)-3\times 2}{2^2-5\times 2+6} = \frac{6\times 1-6}{4-10+6} = \frac{0}{0}$

$\lim_{x\to 2} \frac{6f'(x)-3}{2x-5} = \frac{6f'(2)-3}{2\times 2-5} = \frac{6\times 0-3}{4-5} = \frac{-3}{-1} = 3$

Point 4: 0 or 1

Point 5: 0 or 1

Point 6: 0 or 1

SAMPLE C

$f(x) = x-2,\ 0\le x\le 4$

$\lim_{x\to 2} \frac{6f(x)-3x}{x^2-5x+6} = \lim_{x\to 2} \frac{3(2f(x)-x)}{(x-3)(x-2)} \Rightarrow \frac{2f'(2)}{(x-3)} = 0$

Point 4: 0 or 1

Point 5: 0 or 1

Point 6: 0 or 1

SAMPLE D

$\lim_{x\to 2} \frac{6f(x)-3x}{x^2-5x+6}$

$\lim_{x\to 2} 6f(x)-3x = 0$

$\lim_{x\to 2} x^2-5x+6 = 0$

Since f is differentiable and therefore continous at $x=2$

L'Hospitals Rule applys

$\lim_{x\to 2} \frac{6f'(x)-3}{2x-5} = \frac{-3}{-1} = \boxed{3}$

Point 4: 0 or 1

Point 5: 0 or 1

Point 6: 0 or 1

4.7 L'Hospital's Rule

Calculus

Practice

Find the following. Use L'Hôpital's when possible.

1. $\lim_{x\to 1}\frac{x-1}{x^2-3x+2}$

2. $\lim_{x\to -5}\frac{x^2-2x-35}{x+5}$

3. $\lim_{x\to 0}\frac{4x}{\ln(x+1)}$

4. $\lim_{x\to 0}\frac{x-1}{x^2-3x+2}$

5. $\lim_{x\to 1}\frac{2(x^2-1)}{\ln x^2}$

6. $\frac{d}{dx}\frac{6x^2+x}{\sin(x)}$

7. $\lim_{x\to 0}\frac{2x^2}{e^x-1-x}$

8. $\lim_{x\to 0}\frac{2x^2}{1-\cos(4x)}$

9. $\lim_{x\to 0}\frac{\sqrt{4+x}-2}{x}$

10. $\lim_{x\to -3}\frac{x-1}{x^2+7x+10}$

11. $\lim_{x\to \infty}\frac{e^{2x}}{2x^2}$

12. $\lim_{x\to 0}\frac{\cos x-1}{x^2}$

13. $\frac{d}{dx}\frac{6x^2+x}{x+1}$

14. $\lim_{x\to \infty}\frac{\ln x^2}{\ln(x+4)^3}$

15. $\lim_{x\to -2}\frac{x+2}{x^2+2x-3}$

4.7 L'Hospital's Rule

Test Prep

16. If $f(x) = 2x^3 + 5$, then $\lim_{x \to 0} \frac{f(x) - f(0)}{x^3}$ is

(A) 0 (B) 1 (C) 2 (D) 3 (E) The limit does not exist.

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17. Functions $f, g,$ and h are twice-differentiable functions with $g(3) = h(3) = 5$. The line $y = 5 + \frac{1}{2}(x - 3)$ is tangent to both the graph of g at $x = 3$ and the graph of h at $x = 3$.

a. Find $h'(3)$.

b. Let a be the function given by $a(x) = 2x^3 h(x)$. Write an expression for $a'(x)$. Find $a'(3)$.

c. The function h satisfies $h(x) = \frac{x^2 - 9}{1 - (f(x))^3}$ for $x \neq 3$. It is known that $\lim_{x \to 3} h(x)$ can be evaluated using L'Hospital's Rule. Use $\lim_{x \to 3} h(x) = 5$ to find $f(3)$ and $f'(3)$. Show the work that leads to your answers.